



**UGANDA CHRISTIAN
UNIVERSITY**

A Centre of Excellence in the Heart of Africa

**FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
TRINITY 2026 SEMESTER TEST ONE**

PROGRAM: BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

YEAR: 1

SEMESTER: 2

COURSE CODE: MTH1203

COURSE NAME: PROBABILITY AND STATISTICS

EXAMINATION TYPE: PRACTICAL - 100%

EXAMINATION DATE: SEPTEMBER 2026

TIME ALLOWED: 1 HOUR

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. Attempt all the questions. Every question carries 20 marks.
3. The examination should be executed in Python, and answers to every question should be placed in a new Jupyter Notebook.
4. Every question must include written interpretation: state and explain each result in a Markdown cell, not in code comments alone.
5. Submit a zipped folder containing Jupyter Notebooks and any other exported files.
6. Uganda Christian University or its officials reserve the right of admission or disqualification from this exam.

Question 1

The table below shows the end-of-programme record for 12 first-year IT trainees who completed a practical skills bootcamp. This data is **not** in a file; you must create the DataFrame yourself.

Trainee	Track	Sex	LabHours	Attendance	Score	Result	Certified
Faith	Networking	F	20	90	80	Pass	Yes
Kevin	Software	M	9	62	47	Fail	No
Miriam	Cybersecurity	F	23	96	89	Pass	Yes
Tonny	Software	M	7	57	41	Fail	No
Sandra	Networking	F	19	88	82	Pass	Yes
Peter	Cybersecurity	M	13	72	63	Pass	No
Grace	Software	F	15	79	68	Pass	No
Isaac	Networking	M	11	66	54	Pass	No
Diana	Cybersecurity	F	25	99	93	Pass	Yes
Martin	Software	M	8	59	46	Fail	No
Ruth	Networking	F	10	64	49	Fail	No
Alex	Cybersecurity	M	17	84	74	Pass	Yes

Table 1: End-of-programme record for 12 first-year IT trainees.

- (a) **4 marks**
 Create a pandas DataFrame called `trainees_df` from the table above. State the **expected** number of rows and columns in a Markdown cell **before** running the code, then verify with `.shape` and `.info()`, confirm there are no missing values, and note which columns are categorical.
- (b) **4 marks**
 Compute the mean, median and standard deviation of `LabHours`, `Attendance` and `Score`. Identify the highest- and lowest-scoring trainees, and comment on how spread out the scores are.
- (c) **4 marks**
 For the `Score` column, compute the quartiles `Q1`, `Q2` (the median) and `Q3`, the interquartile range (IQR) and the 90th percentile. Using the $1.5 \times \text{IQR}$ rule, report the outlier boundaries and state whether any trainee is an outlier on `Score`. Then use `.describe()` and compare the mean and median of `Score` to comment on whether the distribution is symmetric or skewed.
- (d) **4 marks**
 Split the trainees into those who passed and those who failed (the `Result` column). Compute the mean `LabHours` and mean `Attendance` for each group, and state which of the two factors shows the larger gap between the pass and fail groups.
- (e) **4 marks**
 Synthesise your findings from (b)–(d) into a data-driven recommendation (120–150 words) to the bootcamp training committee on which single factor best identifies strong trainees. State one limitation of drawing conclusions from only 12 records.

Question 2

The `mpg.csv` dataset contains fuel-economy records for 398 car models. Each record includes the following variables:

Column	Type	Description
<code>mpg</code>	Numeric	Fuel efficiency (miles per gallon)
<code>cylinders</code>	Numeric	Number of cylinders in the engine
<code>displacement</code>	Numeric	Engine displacement (cubic inches)
<code>horsepower</code>	Numeric	Engine horsepower (contains some missing values)
<code>weight</code>	Numeric	Vehicle weight (lbs)
<code>acceleration</code>	Numeric	Time to accelerate 0–60 mph (seconds)
<code>model_year</code>	Numeric	Year of manufacture (e.g. 70 = 1970)
<code>origin</code>	Categorical	Region of origin (usa / japan / europe)
<code>name</code>	Categorical	Car model name

An analyst wishes to summarise the physical and performance profile of the cars and draw conclusions about the wider population from which they were sampled.

(a) **3 marks**

- (i) Load the file into a DataFrame `cars`, then create a working copy `data` that drops all rows containing missing values.
- (ii) Display the first 5 rows and report the shape (rows, columns) **before and after** dropping missing values. State how many rows were lost.

(b) **5 marks**

- (i) For `mpg`, `horsepower` and `weight`, compute the mean, median, mode, standard deviation and interquartile range (IQR). Present the results in a single summary table.
- (ii) Classify each column of `data` as **nominal**, **ordinal**, **discrete** or **continuous**, justifying each choice in one line.

(c) **4 marks**

- (i) Compute the mean `mpg` for each `origin`. State which region is the most, and which is the least, fuel-efficient on average.
- (ii) For each `origin`, compute the interquartile range (IQR) of `mpg`, and use the $1.5 \times \text{IQR}$ rule to report how many cars (if any) are outliers within each region. State which region has the largest spread.

(d) **4 marks**

For `weight`, compute the 10th, 25th, 50th (median), 75th and 90th percentiles. Interpret, in plain words, what the 90th percentile tells you about the cars, and report how many cars are **heavier** than the 90th-percentile value.

(e) **4 marks**

For `horsepower`, compute the mean, standard deviation and range for each `origin`, and present the results as a single grouped summary table. State which region has the highest average horsepower, and which region is the most variable (largest standard deviation).

◀ END ▶